KUKU STORYPULSE: Algorithmic Microdrama Promo Discovery & Diversity Ranking

Production-Grade Editorial Intelligence Platform for Problem 1: Find the Promos (Ads)

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Live Prototype: https://kuku-storypulse.vercel.app | **Repository:** https://github.com/Prajwal9823/kuku-storypulse

1. Executive Summary, Problem Framing, & Strategic Assumptions

Why Problem 1: Microdrama streaming networks (Kuku TV, ReelShort, DramaBox) acquire up to 90% of paying users through 20–60s vertical video ads distributed across TikTok, Instagram Reels, and YouTube Shorts. Traditional manual scrub-and-clip workflows by human editors across 50–100 episodes per season create an acute operational bottleneck, high labor overhead, and missed viral opportunities. Kuku StoryPulse automates the end-to-end extraction, multimodal evaluation, 7-dimension scoring, and diversity-aware reranking of publication-ready promo ads directly from raw screenplays without human fatigue.

Core Computational & Narrative Assumptions:

- **The 3-Second Scroll-Stop Invariant:** Mobile audiences decide whether to swipe within 1.5 to 3.0 seconds. An ad must open on an acute vocal confrontation, shocking revelation, physical slap/threat, or moral outrage. Slow visual establishers cause catastrophic drop-off.
- **The Suspended Curiosity Cut (No Narrative Closure):** A promotional ad is an acquisition engine, not a summary. It must terminate at peak unresolved tension (an interrupted question, weapon drawn, unsealed secret), compelling the viewer to install the app to see the resolution.
- **Cold-Audience Standalone Comprehensibility:** First-time viewers possess zero lore context. Viable promos must establish clear binary stakes (betrayer vs. victim, oppressor vs. undercover duke) without relying on unexplained pronouns or deep historical lore.
- **Portfolio Diversity Over Monotony (MMR Reranking):** Returning the 10 highest raw-scoring clips inevitably creates redundant clusters of the same scene. A production publishing slate must mathematically balance raw strength with narrative and emotional archetype diversity.
- **Strategic Multi-Segment Cross-Episode Assemblies (Max 3 Cuts):** While single continuous clips (20–60s) preserve natural immersion, strategic pairings across distant episodes (e.g., Altar Betrayal in Ep 1 paired with Royal Ballroom Retribution in Ep 42) unlock viral cause-and-effect power.

2. Technical Architecture & The 5-Stage Hybrid Intelligence Pipeline

Kuku StoryPulse completely rejects naive LLM prompt-stuffing (which suffers from hallucinated timestamps, token limits, and nondeterministic drift) in favor of a robust hybrid pipeline combining sub-second AST parsing, narrative graph ontologies, combinatorial window generation, and calibrated heuristic scoring:

- **Stage 1 — Sub-Second Screenplay AST Parser:** Ingests raw screenplays, extracting Scene Headings, exact timestamps (`MM:SS.xx`), Dialogue, Speaker Delivery tags, Physical Actions, Camera Cues, and Sound FX. Handles simultaneous/overlapping dialogue and actions with 100% mathematical precision and zero temporal drift.
- **Stage 2 — Ontological Narrative Intelligence Engine:** Constructs a formal knowledge graph mapping *Character* → *Event* → *Episode* → *Conflict* → *Consequence*. Tracks 8 canonical characters, 6 major narrative arcs, secret identities (Alexander's true ducal heritage), and lore dependencies to calculate standalone comprehensibility penalties for cold viewers.
- **Stage 3 — Multi-Resolution Sliding-Window Candidate Generator:** Evaluates 20.0s to 60.0s candidate windows anchored dynamically on high-energy speech turns, shocking delivery tags, and cliffhanger cut-points. Generates both continuous single-scene clips and cross-episode multi-segment assemblies (explored 6,516 total candidates across the 50 episodes).
- **Stage 4 — 7-Dimension Promo DNA Scoring Formula:** Computes a normalized score (0–100) combining 7 weighted components:
  Promo Score = 0.22(Hook) + 0.22(Cliffhanger) + 0.18(Standalone) + 0.12(Emotion) + 0.10(Conflict) + 0.08(Visual) + 0.08(Novelty) - Penalties
  Diagnostic Deductions: Penalties are applied for slow openings (>3.0s to hook), exposition-heavy dialogue, dangling unanchored pronouns, and premature emotional resolution.
- **Stage 5 — Diversity-Aware Maximal Marginal Relevance (MMR) Reranker:** Formulates slate selection as an optimization problem: **$\mathrm{MMR}(c) = \lambda \cdot \mathrm{Score}(c) - (1 - \lambda) \cdot \max_{s \in S} \mathrm{Similarity}(c, s)$** ($\lambda = 0.55$). Suppresses temporal Jaccard overlaps (>0.15) and distributes promos across 5 archetypes: Betrayal, Revenge, Confrontation, Romance, and Suspense.

3. Empirical Evaluation: Top-Ranked Publishing Slate (50-Episode Pack)

Rank	Title & Core Conflict	Ep(s)	Timestamps	Dur.	Score	Archetype	Scroll-Stop Hook (First 3s) & Unresolved Cliffhanger Cut
#1	**Altar Betrayal & Repudiation**	Ep 2	00:00.00 - 00:23.70	23.7s	**87.2**	Betrayal	**Hook:** Gerald coldly sneers: *'I cannot marry you, Juliana.'* **Cliffhanger:** Cuts on ultimatum: *'Promises change when a better choice appears.'*
#2	**The Truth Indictment**	Ep 36	01:57.50 - 02:33.00	35.5s	**85.3**	Confrontation	**Hook:** Alexander thunders: *'You told them the truth!'* (Furious delivery). **Cliffhanger:** Ominous drone swells to peak; cuts mid-sentence before reply.
#3	**Catherine's Pleading Met with Contempt**	Ep 44	00:25.20 - 00:58.20	33.0s	**85.3**	Revenge	**Hook:** Catherine weeps: *'Alexander... I thought you had died.'* **Cliffhanger:** Alexander turns away coldly; cuts on unanswered desperate plea.
#4	**Cross-Arc Cause & Effect Assembly**	1 + 42	Ep 1: 00:11.30-00:39.00 Ep 42: 00:00.00-00:25.20	52.9s	**84.5**	Assembly (Revenge)	**Hook (Ep 1):** Gerald discards Juliana at church steps in heavy rain. **Cliffhanger (Ep 42):** Wounded soldier reveals Duke's seal; ballroom gasps in terror.
#5	**Thug Ambush & Blade Defense**	Ep 15	00:56.50 - 01:26.00	29.5s	**82.8**	Romance / Action	**Hook:** Thug laughs cruelly: *'Look at her! Crying over weeds!'* **Cliffhanger:** Alexander steps between them with drawn steel; cuts before clash.

4. Quantitative Verification & 10-Point Automated Quality Audit

Every candidate underwent automated mathematical verification across 10 deterministic invariants: **(1) Duration Bounds (20.0s ≤ t ≤ 60.0s):** 100% Passed. **(2) Sub-Second Timestamp Monotonicity:** 100% Passed. **(3) Single-Episode Boundary Containment:** 100% Passed. **(4) Multi-Segment Duration Constraint:** 100% Passed (Cross-episode promo strictly 52.90s). **(5) Segment Count Constraint (≤3 cuts):** 100% Passed. **(6) Dialogue Verbatim Traceability:** 100.0% match against source screenplay text (**Zero Hallucinations Guarantee**). **(7) 7-Dimension Score Invariant (0 ≤ S ≤ 100):** 100% Passed. **(8) Hook Anchor Verification (≤3.0s):** 100% Passed. **(9) Cliffhanger Cut-Point Integrity:** 100% Passed. **(10) Character Attribution Integrity:** 100% Passed across all 8 canonical entities. *Performance:* Full 50-episode pipeline executes in **1.72 seconds**.

5. Honest Failure Analysis & Algorithmic Edge Cases (Where Discovery Fails)

- **High Drama with Heavy Lore Prerequisites (Ep 25: 00:35–01:15, Raw Score 78.5):** Juliana discovers Catherine's arsenic vial. While emotionally intense for existing fans, cold TikTok viewers lack context for who the 'Dowager Duchess' is, leading to confusion. *Mitigation:* Implemented Lore Penalty that deducts 15 points if character motivation relies on more than 2 unresolved entity premises.
- **Late-Peaking High Climax with Quiet Establishes (Ep 10: 00:00–00:42, Raw Score 72.0):** Alexander's bloody wound confession peaks dramatically at 00:35, but the opening 6.5s consists of silent sponge washing. Mobile users swipe away before the climax. *Mitigation:* Implemented Dynamic Hook Trimming that nudges the candidate window +5.5s forward to open directly on spoken dialogue.
- **Premature Catharsis & Conflict Resolution (Ep 14: 00:45–01:22, Raw Score 69.0):** Alexander bandaging Juliana's torn hands has exceptional romantic tension, but concludes with soothing reassurance and smiles. Psychological closure removes urgency to watch the next episode. *Mitigation:* Cliffhanger detector penalizes segments ending on de-escalation sentiments by 20 points.
- **Extreme Cross-Episode Disorientation (30-Episode Leap Cut):** Splicing an early ballroom scene with a late-season dungeon interrogation caused severe viewer disorientation due to unbridged character state jumps. *Mitigation:* Restricted multi-segment candidates to a maximum of 2 arcs and enforced a transition coherence score.

6. Production Scalability & Operational Blueprint Across Every Kuku Show

1. Universal Screenplay Portability: The architecture contains zero hardcoded show names, episode counts, or fixed character registries. Ingesting any new season ZIP file dynamically reconstructs the narrative graph and discovers top promos in <15 seconds.
2. Multi-Tier AI Resilience (Deterministic + Cloud LLM): Operates completely offline with 100% deterministic mathematical scoring, while supporting pluggable LLM reasoning (OpenAI, Gemini, Groq, Ollama) via a unified provider interface for human-like editorial explanations.
3. Real-Time Human-in-the-Loop Studio: The live Next.js editorial suite allows video editors to drag timeline handles to trim sub-second boundaries, inspect 7-dimension Promo DNA radar charts, review automated QC badges, and export final broadcast packages to CSV, JSON, and PDF.